DATA STRUCTURE AND ALGORITHM

Programming Practice Session

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Instructor Information

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Regulations and Grading Policy

- 6 practical sessions and 1 practical exam.
- Sign the attendance sheet in each session.
- Turn off or switch your phone to silent mode in class.
- Avoid being late or absent from class.
- Emails or messages requesting grades will not be answered.
- Grades reflect the outcomes of the learning process.
- Check email, courses, group regularly.

Regulations and Grading Policy

- The practical session accounts for 30% of total grade.
- Each session includes: attendance test, end-of-class test and homework assignments.
- The practical exam accounts for 70% of the practical.
- Bonus points (up to 1 point) will be added to the practical exam score and/or may determine the final rounding of the grade.
- C++ required.

Regulations and Grading Policy

- Code must include comments
- Use appropriate data types and meaningful names for variables and functions.
- Submit assignments according to the requirements, with filename format: AssignmentOrder-StudentID-FullName, e.g., “[Lab1]23529999-NguyenVanA.cpp”
- Paper tests must include: Full Name, Student ID and Session.

1st Session

Big-O Notation (Complexity)
Searching
Sorting

Quiz

1. Which of the following is a linear data structure?

A. Array

B. Linked List

C. Binary Tree

D. Graph

Quiz

2. What is the worst-case time complexity of Linear Search?

A. O(1)

B. O(log n)

C. O(n)

D. O(n log n)

Quiz

3. Which sorting algorithm is NOT comparison-based?

A. Merge Sort

B. Quick Sort

C. Counting Sort

D. Heap Sort

Quiz

4. What is the average-case time complexity of Binary Search?

A. O(1)

B. O(log n)

C. O(n)

D. O(n log n)

Quiz

5. In Big-O notation, which describes the fastest growing function?

A. O(n)

B. O(n log n)

C. O(2^n)

D. O(n^2)

Quiz

6. Which data structure works on the principle of FIFO (First In First Out)?

A. Stack

B. Queue

C. Tree

D. Graph

Quiz

7. What is the best-case time complexity of Insertion Sort?

A. O(n)

B. O(n log n)

C. O(n^2)

D. O(1)

Quiz

8. The space complexity of an algorithm refers to:

A. Maximum number of instructions executed

B. Amount of memory required

C. Time required for execution

D. Number of variables declared

Quiz

9. A _____ is a Last In First Out (LIFO) data structure.

10. Binary Search requires the input array to be _____.

11. The algorithmic complexity that describes constant time is _____.

12. In Quick Sort, the element used to divide the array is called the _____.

Quiz

13. int linearSearch(int arr[], int n, int target) {

```
    for (int i = 0; i < (a); i++) {
        if ((b)== target) {
            return (c); }}
    return -1;}
```

Quiz 14.

```
int binarySearch(int arr[], int n, int target) {
    int left = 0, right = n - 1;
    while ((a) <= (b)) {
        int mid = (left + right) / 2;
        if (arr[mid] == target)        return (c);
        else if (arr[mid] < target)    left = (d) + 1;
        else                           right = (e) - 1;
    }
    return -1;}
```

Quiz

```
15. void bubbleSort(int arr[], int n) {
    for (int i = 0; i < (a); i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] (b) arr[j + 1]) {
                (c);
            }
        }
    }}
```

Assignment 1

1. Re-implement (C++) learned algorithms:
   - Search: Linear Search, Binary Search.
   - Sorting: Selection Sort, Insertion Sort, Bubble Sort, Quick Sort, Heap Sort, Radix Sort.
2. Re-implement (C++) additional algorithms:
   - Search: Jump Search, Ternary Search, Interpolation Search, Exponential Search.
   - Sorting: Counting Sort, Bucket Sort, Merge Sort, Tim Sort.
3. Report: algorithm idea, characteristics, code, visualization (print each step result).
4. Submit: a <studentID>.zip file includes codes.cpp and report.pdf (should be in latex format).